

Basic Statistics

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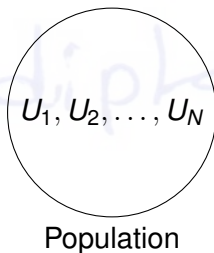
- 1 Sampling Theory
 - Sampling from Finite Population
 - Sampling from Theoretical Population
 - Sampling from Normal Population
 - Large Sampling

Chapter 7: Sampling Theory

Sudipta Das

Performing the study of a **population** with the help of a **sample**

- Population: - A collection/aggregate of all the units/individuals possessing a common characteristics.



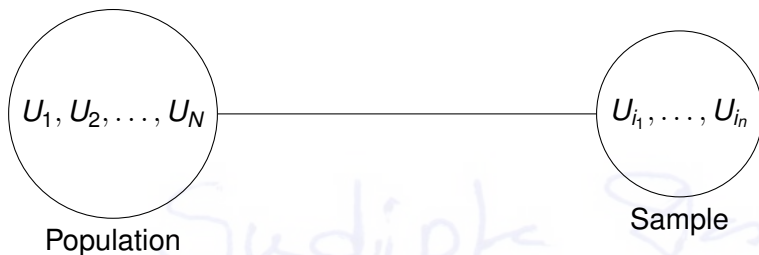
- Let's consider a population of size N units, say $(U_1, U_2, \dots, U_N)$.
 - N is called population size
- Let Y be the character under study, having values $Y_1^{(p)}, Y_2^{(p)}, \dots, Y_N^{(p)}$ for those N units.

- Three types of population
 - 1 Real and finite
 - the population of students studying data analytic courses
 - 2 Infinite or hypothetical
 - the population of all the tosses that can be made with a coin
 - the population of all the stars in our galaxy
 - 3 Theoretical in nature
 - the binomial population or normal population etc.
- A population can be studied either by complete enumeration (census) or by choosing some units from population (sampling)

Census or Complete enumeration:

- By census, we mean inspection/enumeration/study of **all the units** belonging to a population.
 - If we undertake a census, we have knowledge of $Y_1^{(p)}, Y_2^{(p)}, \dots, Y_N^{(p)}$ individually.
- Problem with census
 - Naturally, census possible for a finite population only.
 - For infinite population census is impossible
 - For finite population there may be restriction in time, cost or space.
 - Destructive sampling: to study the length of life (in hours) of the bulbs produced by a company.

Sampling V



Sampling:

- This consists of selection/inspection of **only a few units** of population
- The selected/inspected units as a group is called a sample.

- Types of sampling
 - 1 Non-probabilistic (judgmental) sampling
 - Not helpful
 - 2 Probabilistic sampling
 - It helps in statistical analysis and decision making.
- Based on the study of probability, the present study of *sampling* marks the beginning the learning of statistics beyond the descriptive phase.

Sampling from Finite Population I

- Simple Random Sampling (SRS):
 - Suppose we draw n units from the population of N units, as our sample.
 - n is called sample size
 - The sample will be called simple random sample, if **every possible sample of size n has the same probability of being selected**

Sampling from Finite Population II

- Three types of Simple Random Sampling (SRS)
 - 1 SRS with replacement (SRSWR)
 - Application:- Bagging
 - 2 SRS without replacement (SRSWOR)
 - Application:- Subsampling
 - 3 SRS without replacement unordered (SRSWOR unordered)

Sampling from Finite Population III

- Simple Random Sampling with replacement (SRSWR)
 - In this case, after the first unit is selected at random from all the N units, it is replaced to the population, and, the second unit is drawn, again at random, from all the N units of the population.
 - This procedure is repeated n times to get an SRS of size n drawn with replacement from the population
- SRSWR for a population of size N ,
 - total number of possible sample of size n is

$$N^n$$

- and selection probability of each sample is $1/N^n$

Sampling from Finite Population IV

- Simple Random Sampling without replacement (SRSWOR)
 - Here after the first unit is selected at random from all the N units, it is not replaced to the population, and, the second unit is drawn, again at random, from the remaining $N - 1$ units of the population.
 - This procedure is repeated n times to get an SRS of size n drawn without replacement from the population
- SRSWOR for a population of size N ,
 - total number of possible sample of size n is

$$N(N - 1)(N - 2) \cdots (N - n + 1) = {}^N P_n$$

- and selection probability of each sample is $\frac{1}{{}^N P_n}$

Sampling from Finite Population V

- Simple Random Sampling without replacement unordered (SRSWOR unordered)
 - If the order of appearance of the units in the sample can be ignored, then we have the unordered SRSWOR procedure
- SRSWOR unordered for a population of size N ,
 - total number of possible sample of size n is

$$\frac{N(N-1)(N-2)\cdots(N-n+1)}{1 \times 2 \times \cdots \times n} = {}^N C_n$$

- and selection probability of each sample is $\frac{1}{{}^N C_n}$

Sampling from Finite Population VI

- Example: Population $\{U_1, U_2, U_3, U_4\}$ $N=4$ with $n = 2$

SRSWR

$$\left. \begin{array}{cccc} U_1, U_1 & U_1, U_2 & U_1, U_3 & U_1, U_4 \\ U_2, U_1 & U_2, U_2 & U_2, U_3 & U_2, U_4 \\ U_3, U_1 & U_3, U_2 & U_3, U_3 & U_3, U_4 \\ U_4, U_1 & U_4, U_2 & U_4, U_3 & U_4, U_4 \end{array} \right\} 16$$

SRSWOR(ordered)

$$\left. \begin{array}{cccc} -- & U_1, U_2 & U_1, U_3 & U_1, U_4 \\ U_2, U_1 & -- & U_2, U_3 & U_2, U_4 \\ U_3, U_1 & U_3, U_2 & -- & U_3, U_4 \\ U_4, U_1 & U_4, U_2 & U_4, U_3 & -- \end{array} \right\} 12$$

SRSWOR(unordered)

$$\left. \begin{array}{cccc} -- & U_1, U_2 & U_1, U_3 & U_1, U_4 \\ -- & -- & U_2, U_3 & U_2, U_4 \\ -- & -- & -- & U_3, U_4 \\ -- & -- & -- & -- \end{array} \right\} 6$$

- Objective: To know about a parameter of the population from the collected sample

Sampling from Finite Population VII

- Parameter: A parameter is a real-valued measurable function of the population values, say

$$\theta = f(Y_1^{(p)}, Y_2^{(p)}, \dots, Y_N^{(p)}),$$

where $(Y_1^{(p)}, Y_2^{(p)}, \dots, Y_N^{(p)})$ are the population values of a finite population

Sampling from Finite Population VIII

- Population parameters, e.g.,

① Population mean: $\mu = \frac{1}{N} \sum_{i=1}^N Y_i^{(p)}$

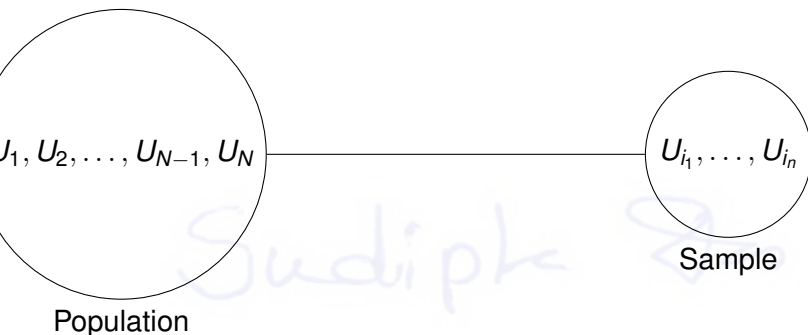
② Population variance: $\sigma^2 = \frac{1}{N} \sum_{i=1}^N \left(Y_i^{(p)} - \mu \right)^2$

- ③ Population proportion of some character:

$$p = \frac{X^{(p)}}{N},$$

where $X^{(p)}$ = number of units possessing the character in the population

Sampling from Finite Population IX



- Population values $Y(U_1) = Y_1^{(p)}, Y(U_2) = Y_2^{(p)}, \dots, Y(U_N) = Y_N^{(p)}$
- Sample values $Y(U_{i_1}) = Y_1, Y(U_{i_2}) = Y_2, \dots, Y(U_{i_n}) = Y_n$, where $\{i_1, \dots, i_n\}$ are integers from $\{1, 2, \dots, N\}$
 - Note that the sample values are RANDOM VARIABLE

Sampling from Finite Population X

- **Statistic:** A statistic is a real valued measurable function of the sample values $(Y_1, Y_2, \dots, Y_n)$, say

$$T = T(Y_1, Y_2, \dots, Y_n),$$

For example

- Sample mean $= \bar{Y} = \frac{Y_1 + Y_2 + \dots + Y_n}{n} = \frac{1}{n} \sum_{i=1}^n Y_i$
- Sample variance $= s_Y^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$
- Sample proportion of some character $= \hat{p} = \frac{X}{n}$, where X = number of units possessing the character in the sample
- **Note:**
 - Any statistics, being a function of n random variables $(Y_1, Y_2, \dots, Y_n)$, is itself a random variable, having a certain probability distribution.

Sampling from Finite Population XI

- Sampling Distribution of a statistic:
 - is the probability distribution of the statistic (which is a random variable) for repeated sampling each of size n from the same population.

Sampling from Finite Population XII

- Expectation and Standard error of a statistic T with sampling distribution given by a p.m.f $f(t)$

- Expectation of T

$$\mu_T = E(T) = \sum_t t \times f(t),$$

which is the mean of the sampling distribution

- Standard error of T

$$\sigma_T = SE(T) = +\sqrt{\text{var}(T)},$$

where $\text{var}(T) = E(T - \mu_T)^2 = \sum_t (t - \mu_T)^2 \times f(t).$

Study on Sample Mean

Problem:

- A population has 4 values ($Y_1^{(P)} = 1, Y_2^{(P)} = 2, Y_3^{(P)} = 4, Y_4^{(P)} = 5$) and we draw a sample of size 2 from this population. Derive the sampling distribution of the sample mean. Also find the expectation and the standard error of the sample mean.

Sampling from Finite Population XIV

Solution steps:

- Population size, $N = 4$ and Sample size, $n = 2$,
- Population mean, $\mu = \frac{1}{N} \sum_{i=1}^N Y_i^{(P)} = 3.0$
- Population variance, $\sigma^2 = \frac{1}{N} \sum_{i=1}^N \left(Y_i^{(P)}\right)^2 - \mu^2 = 2.50$
- Population standard deviation, $\sigma = \sqrt{2.50}$

Sampling from Finite Population XV

Table: Possible sample and the sample mean ($\bar{Y}$) in SRSWR

Sample No	Sample Values	Prob	Sample Mean	Sample No	Sample Values	Prob	Sample Mean
1	(1,1)	1/16	1.0	9	(4,1)	1/16	2.5
2	(1,2)	1/16	1.5	10	(4,2)	1/16	3.0
3	(1,4)	1/16	2.5	11	(4,4)	1/16	4.0
4	(1,5)	1/16	3.0	12	(4,5)	1/16	4.5
5	(2,1)	1/16	1.5	13	(5,1)	1/16	3.0
6	(2,2)	1/16	2.0	14	(5,2)	1/16	3.5
7	(2,4)	1/16	3.0	15	(5,4)	1/16	4.5
8	(2,5)	1/16	3.5	16	(5,5)	1/16	5.0

Table: Sampling distribution of the sample mean ($\bar{Y}$) in SRSWR

$\bar{Y}$	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	Total
Prob	1/16	2/16	1/16	2/16	4/16	2/16	1/16	2/16	1/16	1

Sampling from Finite Population XVI

- Mean of the sample mean ($\bar{Y}$), in SRSWR:

$$\text{Expectation of } \bar{Y} = E[\bar{Y}] = 3.0$$

- Variance of the sample mean ($\bar{Y}$), in SRSWR:

$$\text{Variance of } \bar{Y} = E[\bar{Y}^2] - E^2[\bar{Y}] = 1.25$$

- Standard error of the sample mean ($\bar{Y}$), in SRSWR:

$$\text{Square root of Variance of } \bar{Y} = SE[\bar{Y}] = \sqrt{1.25}$$

- Note that

- $E[\bar{Y}] = \mu = \text{population mean} = 3.0$
- $SE[\bar{Y}] = \frac{\sigma}{\sqrt{n}} = \frac{\text{population s.d}}{\sqrt{\text{sample size}}} = \frac{\sqrt{2.5}}{\sqrt{2}}$

Sampling from Finite Population XVII

Table: Possible sample and the sample mean ($\bar{Y}$) in SRSWOR

Sample No	Sample Values	Prob	Sample Mean
1	(1,2)	1/6	1.5
2	(1,4)	1/6	2.5
3	(1,5)	1/6	3.0
4	(2,4)	1/6	3.0
5	(2,5)	1/6	3.5
6	(4,5)	1/6	4.5

Table: Sampling distribution of the sample mean ($\bar{Y}$) in SRSWOR

$\bar{Y}$	1.5	2.5	3.0	3.5	4.5	Total
Prob	1/6	1/6	2/6	1/6	1/6	1

Sampling from Finite Population XVIII

- Mean of the sample mean ($\bar{Y}$), in SRSWOR:

$$\text{Expectation of } \bar{Y} = E[\bar{Y}] = 3.0$$

- Variance of the sample mean ($\bar{Y}$), in SRSWOR:

$$\text{Variance of } \bar{Y} = E[\bar{Y}^2] - E^2[\bar{Y}] = 5/6$$

- Standard error of the sample mean ($\bar{Y}$), in SRSWOR:

$$\text{Square root of Variance of } \bar{Y} = SE[\bar{Y}] = \sqrt{5/6}$$

- Note that

- $E[\bar{Y}] = \mu = 3.0$

- $SE[\bar{Y}] = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}} = \frac{\sqrt{2.5}}{\sqrt{2}} \times \sqrt{\frac{4-2}{4-1}}$

Sampling from Finite Population XIX

Theorem

- Consider a finite population of N units having mean μ and variance σ^2 . Suppose we draw a simple random sample of size n from the above population. If $\bar{Y}$ denotes the sample mean, then
 - For SRSWR,
 - $E[\bar{Y}] = \mu$
 - $Var(\bar{Y}) = \frac{\sigma^2}{n} \Rightarrow SE[\bar{Y}] = \frac{\sigma}{\sqrt{n}}$
 - For SRSWOR,
 - $E[\bar{Y}] = \mu$
 - $Var(\bar{Y}) = \frac{\sigma^2}{n} \frac{N-n}{N-1} \Rightarrow SE[\bar{Y}] = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$

Sampling from Finite Population XX

Remarks

- The factor $\frac{N-n}{N-1}$ is called the **finite population correction factor**.
- If $n > 1$, then $SE(\bar{Y}|SRSWOR) < SE(\bar{Y}|SRSWR)$
- For a fixed n , as $N \rightarrow \infty$, the $SE(\bar{Y}|SRSWOR) \rightarrow SE(\bar{Y}|SRSWR)$
 - As sampling from a practically infinite population, it is immaterial whether the units already drawn are returned or not before drawing the next unit.

Sampling from Finite Population XXI

Sketch of proof (SRSWR)

- Support of the i^{th} sample unit (Y_i) is $\{Y_1^{(P)}, Y_2^{(P)}, \dots, Y_k^{(P)}, \dots, Y_l^{(P)}, \dots, Y_N^{(P)}\}$
- $P(Y_i = Y_k^{(P)}) = \frac{1}{N}$
- $P(Y_i = Y_k^{(P)}, Y_j = Y_l^{(P)}) = \frac{1}{N^2}$
- $E(Y_i) = \sum_{k=1}^N Y_k^{(P)} P(Y_i = Y_k^{(P)}) = \frac{1}{N} \sum_{k=1}^N Y_k^{(P)} = \mu$
- $Var(Y_i) = \sum_{k=1}^N (Y_k^{(P)} - \mu)^2 P(Y_i = Y_k^{(P)}) = \frac{1}{N} \sum_{k=1}^N (Y_k^{(P)} - \mu)^2 = \sigma^2$
- $Cov(Y_i, Y_j) = 0$
- $E[\bar{Y}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n} \left[\sum_{i=1}^n EY_i\right] = \frac{1}{n} \times n\mu = \mu$
- $Var[\bar{Y}] = Var\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n^2} \left[\sum_{i=1}^n Var(Y_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n Cov(Y_i, Y_j)\right] = \frac{1}{n^2} [n\sigma^2] = \frac{\sigma^2}{n}$

Sampling from Finite Population XXII

Sketch of proof (SRSWOR)

• Support of the i^{th} sample unit (Y_i) is $\{Y_1^{(P)}, Y_2^{(P)}, \dots, Y_k^{(P)}, \dots, Y_l^{(P)}, \dots, Y_N^{(P)}\}$

• $P(Y_i = Y_k^{(P)}) = \frac{N-1 P_{n-1}}{N P_n} = \frac{1}{N}$

• $P(Y_i = Y_k^{(P)}, Y_j = Y_l^{(P)}) = \frac{N-2 P_{n-2}}{N P_n} = \frac{1}{N(N-1)}$

• $E(Y_i) = \sum_{k=1}^N Y_k^{(P)} P(Y_i = Y_k^{(P)}) = \frac{1}{N} \sum_{k=1}^N Y_k^{(P)} = \mu$

• $Var(Y_i) = \sum_{k=1}^N (Y_k^{(P)} - \mu)^2 P(Y_i = Y_k^{(P)}) = \frac{1}{N} \sum_{k=1}^N (Y_k^{(P)} - \mu)^2 = \sigma^2$

• $Cov(Y_i, Y_j) = \sum_{k=1}^N \sum_{\substack{l=1 \\ k \neq l}}^N (Y_k^{(P)} - \mu)(Y_l^{(P)} - \mu) P(Y_i = Y_k^{(P)}, Y_j = Y_l^{(P)}) = -\frac{\sigma^2}{N-1}$

• $E[\bar{Y}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n} \left[\sum_{i=1}^n EY_i\right] = \frac{1}{n} \times n\mu = \mu$

• $Var[\bar{Y}] = Var\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n^2} \left[\sum_{i=1}^n Var(Y_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n Cov(Y_i, Y_j)\right] = \frac{1}{n^2} \left[n\sigma^2 + 2^n C_2 \left(\frac{-\sigma^2}{N-1}\right)\right] = \frac{\sigma^2}{n} \frac{N-n}{N-1}$

Study on Sample Proportion

- Suppose we interested in estimating the population proportion of some character (e.g. proportion of smokers in a city), say p .
- We draw an SRS of size n and let $\hat{p}$ be the corresponding sample proportion
 - Sample Proportion, $\hat{p}$ is a statistic
 - We are interested to know the expectation and standard error of $\hat{p}$

Sampling from Finite Population XXIV

- Define the N population values as

$$Y_i^{(P)} = \begin{cases} 1, & \text{if the } i^{\text{th}} \text{ unit possesses the character} \\ 0, & \text{otherwise} \end{cases}$$

- Thus

- Population mean $= \mu = \frac{1}{N} \sum_{i=1}^N Y_i^{(P)} = \frac{X^{(P)}}{N} = p = \text{Population proportion}$

- Population variance $= \sigma^2 = \frac{1}{N} \sum_{i=1}^N \left(Y_i^{(P)} \right)^2 - p^2 = p(1 - p)$

- Therefore, the n sample values are

$$Y_i = \begin{cases} 1, & \text{if the } i^{\text{th}} \text{ unit possesses the character} \\ 0, & \text{otherwise} \end{cases}$$

- Sample mean $= \bar{y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{X}{n} = \hat{p} = \text{Sample proportion}$

Sampling from Finite Population XXV

Theorem

- If $\hat{p}$ be the sample proportion for a simple random sample of size n drawn from a population of size N having population proportion p , then
 - For SRSWR,
 - $E[\hat{p}] = p$
 - $Var(\hat{p}) = \frac{p(1-p)}{n} \Rightarrow SE[\bar{Y}] = \sqrt{\frac{p(1-p)}{n}}$
 - For SRSWOR,
 - $E[\hat{p}] = p$
 - $Var(\hat{p}) = \frac{p(1-p)}{n} \times \frac{N-n}{N-1} \Rightarrow SE[\bar{Y}] = \sqrt{\frac{p(1-p)}{n} \times \frac{N-n}{N-1}}$

Sampling from Theoretical Population I

- (I.I.D.) Random Sampling: A number of random variables is selected from a population of **identical** random variables and the random variables are selected **independently** one from another
 - n , (sample size) is the number of selected random variables
- Note that
 - Any function T of observable random variables $X_1, \dots, X_n$ that does not depend on any unknown parameters is called a statistic.
 - The probability distribution of the sample statistic T is called the sampling distribution of T

Sampling from Theoretical Population II

Theorem:

- Let $X_1, \dots, X_n$ be a random sample of size n from a population with mean μ and variance σ^2 . Then $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean having

$$E(\bar{X}) = \mu$$

and

$$Var(\bar{X}) = \frac{\sigma^2}{n} \Rightarrow SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

- Note: The sample means become more and more reliable as an estimate of μ as the sample size is increased,

Sampling from Normal Population I

- **Normal Population:** The random variables (units) in the population are normally distributed.

Normal Random Variable

- A random variable X is said to be normal if its probability density function is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2},$$

where $-\infty < \mu < \infty$ and $0 < \sigma < \infty$.

- Notation: $X \sim N(\mu, \sigma)$
- μ is called the expectation/mean of X , i.e., $E[X] = \mu$
- σ^2 is called the variance of X , i.e., $E[X - \mu]^2 = \sigma^2$
- Moment Generating Function (mgf): $M_X(t) = E(e^{tX}) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

Standardized Normal Random Variable

- If X is normal random variable with mean μ and variance σ^2 , then the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is said to be standardized normal random variable.

- We denote this by $Z \sim N(0, 1)$

Sampling from Normal Population IV

- Probability density function of a standardized normal random variable Z is given by

$$f_Z(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

- Mean: $\mu_Z = 0$
- Variance: $\sigma_Z^2 = 1$
- MGF: $M_Z(t) = e^{\frac{1}{2}t^2}$

Distribution of Sample Mean ($\bar{X}$)

- *Theorem 1:* Let $\{X_1, \dots, X_n\}$ be a random sample of size n , drawn from a normal population with mean μ and variance σ^2 . Then

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is Normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.

- Thus,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = Z \sim N(0, 1)$$

χ^2 - distribution

- If Z_i s are n independent standardized normal variables, then the random variable

$$K = \sum_{i=1}^n Z_i^2$$

is said to have a Chi-square distribution with n degrees of freedom.

- We denote this by $K \sim \chi_n^2$

Sampling from Normal Population VII

- Probability density function of a (centralized) χ^2 random variable K with degree of freedoms n , is given by

$$f_K(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}.$$

- Mean: $\mu_K = n$
- Variance: $\sigma_K^2 = 2n$
- MGF: $M_K(t) = (1 - 2t)^{-n/2}$, for $t < \frac{1}{2}$

Sampling from Normal Population VIII

Some observations

- Suppose the random sample $\{X_1, \dots, X_n\}$ is drawn from a normal population with mean μ and variance σ^2 . Equivalently, $X_i \sim N(\mu, \sigma^2)$. Then

$$Z_i = (X_i - \mu)/\sigma, \text{ for } i = 1, \dots, n$$

are independent standard normal random variables.

- The square of standard normal random variables

$$Z_i^2 = \left(\frac{X_i - \mu}{\sigma} \right)^2 \text{ for } i = 1, \dots, n$$

has a χ^2 -distribution with 1 degrees of freedom.

- MGF: $M_{Z_i^2}(t) = (1 - 2t)^{-1/2}$, for $t < \frac{1}{2}$

Sampling from Normal Population IX

- *Theorem 2:* Suppose the random sample $\{X_1, \dots, X_n\}$ is drawn from a $N(\mu, \sigma^2)$ distributed population. Then $Z_i = (X_i - \mu)/\sigma, i = 1, \dots, n$ are independent standard normal random variables. Thus the random variable

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2$$

has a χ^2 -distribution with n degrees of freedom.

- MGF: $M(t) = (1 - 2t)^{-n/2}$, for $t < \frac{1}{2}$

Sampling from Normal Population X

Distribution of Sample Variance (S^2)

- *Theorem 3:* If $\{X_1, \dots, X_n\}$ is a random sample from a normal population with the mean μ and variance σ^2 , then the random variable

$$\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \frac{(n-1)S^2}{\sigma^2}$$

has a χ^2 -distribution with $n - 1$ degrees of freedom, where

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The sample mean $\bar{X}$ and sample variance S^2 are independent, also.

- Sketch of proof:

$$\underbrace{\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2}_n = \sum_{i=1}^n \left[\frac{(X_i - \bar{X}) + (\bar{X} - \mu)}{\sigma} \right]^2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 + n \left(\frac{\bar{X} - \mu}{\sigma} \right)^2 = \underbrace{\frac{(n-1)S^2}{\sigma^2}}_{n-1} + \underbrace{\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2}_1$$

***t*-distribution**

- If Y and Z are two independent random variables, such that $Y \sim \chi_n^2$ and $Z \sim N(0, 1)$, then the random variable

$$T = \frac{Z}{\sqrt{Y/n}}$$

is said to have a (Student) *t*-distribution with n degrees of freedom.

- We denote this by $T \sim t_n$

Sampling from Normal Population XII

- Probability density function of a T random variable with degree of freedom n , is given by

$$f_T(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}.$$

- Mean: $\mu_T = 0$
- Variance: $\sigma_T^2 = \begin{cases} \frac{n}{n-2}, & \text{for } n > 2 \\ 1, & \text{for } 1 < n \leq 2 \end{cases}$
- MGF: $M_T(t)$ is undefined

Distribution of Sample Mean standardized by Sample Variance

- *Theorem 4:* If $\bar{X}$ and S^2 are the mean and the variance of a random sample of size n , drawn from a normal population with the mean μ and variance σ^2 , then the statistic (random variable)

$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t -distribution with $n - 1$ degrees of freedom

F distribution

- If U and V are two independent chi-square random variables with n_1 and n_2 degrees of freedom, respectively. Then the random variable

$$F = \frac{U/n_1}{V/n_2}$$

is said to have an F-distribution with (n_1, n_2) degrees of freedom.

- We denote this by $F \sim F_{n_1, n_2}$

Sampling from Normal Population XV

- Probability density function of a (centralized) F random variable with degrees of freedom n_1 and n_2 , is given by

$$f_F(x) = \frac{1}{x B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \sqrt{\frac{(n_1 x)^{n_1} n_2^{n_2}}{(n_1 x + n_2)^{n_1 + n_2}}}.$$

- Mean: $\frac{n_2}{n_2 - 2}$ for $n_2 > 2$
- Variance: $\frac{2n_2^2(n_1 + n_2 - 2)}{n_1(n_2 - 2)^2(n_2 - 4)}$ for $n_2 > 4$
- MGF: $M_F(t)$ does not exist

Distribution of Ratio of Sample Variances

- *Theorem 5:* Let two independent random samples of size n_1 and n_2 be drawn from two normal populations with variances σ_1^2 , and σ_2^2 , respectively. If the variances of the random samples are given by S_1^2 and S_2^2 , respectively, then the statistic (random variable)

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

has a F -distribution with $(n_1 - 1), (n_2 - 1)$ degrees of freedom.

- *Corollary:* Under the equality assumption of two population variances (i.e., $\sigma_1^2 = \sigma_2^2$) the statistic (random variable)

$$F = \frac{S_1^2}{S_2^2}$$

has a F -distribution with $(n_1 - 1), (n_2 - 1)$ degrees of freedom

Distribution of Large Sample Mean ($\bar{X}$)

- *Central Limit Theorem (CLT)*: Suppose $\{X_1, \dots, X_n\}$, a random sample of size n , is drawn from a population (*not necessarily normal*) with mean μ and finite variance σ^2 . Then the standardized sample mean

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \text{ as } n \rightarrow \infty.$$